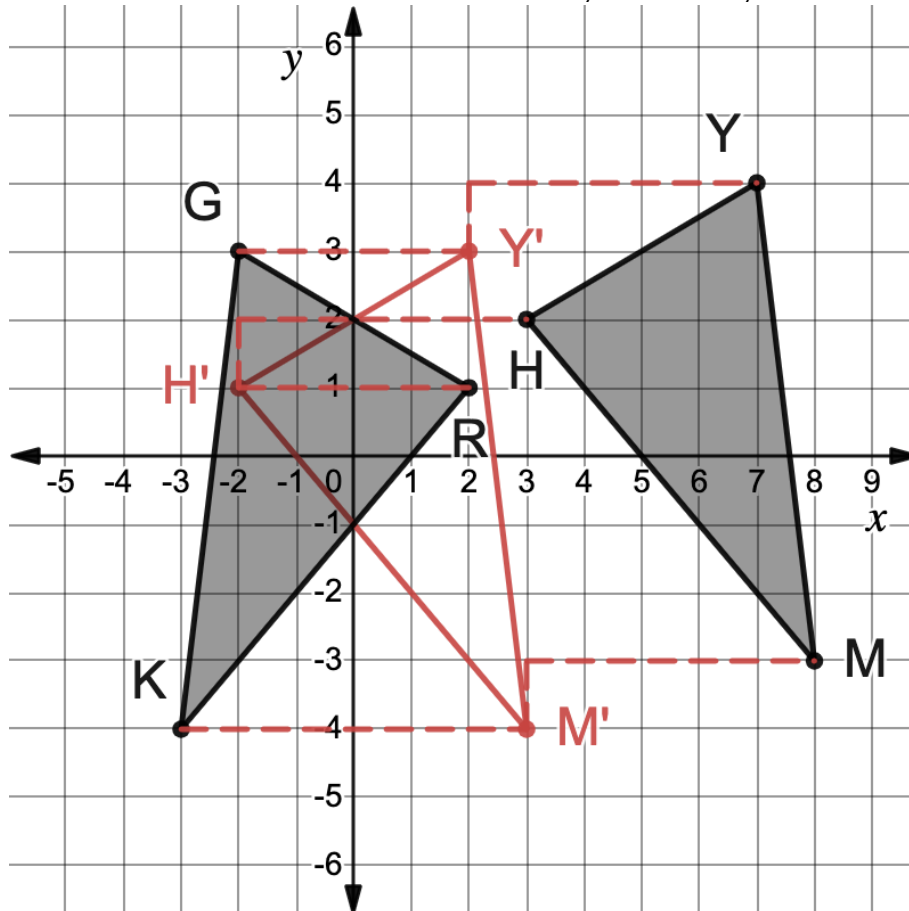


Triangles GRK and MHY are shown. Assume $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$.



1. Explain how to use geometric tools to verify that $\triangle HYM \cong \triangle RGK$.

Use either a compass or a ruler to show that $GR = HY$, $RK = HM$, and $KG = YM$. Then the triangles are congruent by SSS.

2. Write a sequence of transformations that will map $\triangle HYM$ onto $\triangle RGK$.

Answers may vary.

Translate $\triangle HYM$ left 6 and down 1, then reflect across the y -axis.

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'Y'M'$ the result of the first transformation.
4. Map the vertices of $\triangle HYM$ to $\triangle H'Y'M'$. $(x, y) \rightarrow (x - 5, y - 1)$
5. Map the vertices of $\triangle H'Y'M'$ to $\triangle RGK$. $(x, y) \rightarrow (-x, y)$

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HYM$ to $\triangle H'Y'M'$ and then from $\triangle H'Y'M'$ to $\triangle \underline{RGK}$ shows that point H' coincides with point $\underline{R}$, point Y' coincides with point $\underline{G}$, and point M' coincides with point $\underline{K}$.

7. Because the definition of congruence in terms of rigid motions preserves distance,
it can be stated that $\triangle HYM \cong \triangle \underline{RGK}$.

8. Since it was assumed that $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$ then SSS is sufficient to show the triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☒ $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, $\angle GRK \cong \angle YHM$
- ☐ $\overline{GR} \cong \overline{HY}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$
- ☒ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RKG \cong \angle YMH$
- ☐ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{GR} \cong \overline{HY}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☒ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{GR} \cong \overline{HY}$
- ☐ $\angle GRK \cong \angle YHM$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☒ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☒ $\angle RKG \cong \angle YMH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{GR} \cong \overline{HY}$